YongHwi Song

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EDUCATION

Mar. 2023 ~ Feb. 2027 (Expected)

Chungbuk National University — Cheongju, Korea
School of Information and Communication Engineering

Bachelor Student
GPA: 3.83 / 4.5

RESEARCH INTERESTS

- World Models and World-Action Models
- Model-Based Planning and Reinforcement Learning
- Efficient and Reliable World Models
- World Models for Embodied and Language Agents

RESEARCH EXPERIENCES

- Research Intern at Artificial Intelligence Laboratory (AI Lab), Chungbuk National University, Korea (Jun. 2026 ~ Present) / Advisor: Prof. Keon Myung Lee / Conducting research on efficient world and world-action models and on world models for tool-using language agents.
  - Showed that MPPI planners exploit deterministic rounding error in low-bit TD-MPC2 world models; training-free dithering removed a median 95% of the success drop on Meta-World.
  - Found that latent world-model imagination blurs near misses: on DINO-WM PushT, success-vs-near-miss AUROC fell from 0.953 to 0.853, while far misses stayed separable.
  - Showed that a frozen decision model, JEV, works as an LLM agent's world model without task training, reaching 93.4% success in TextWorld vs. 79.5% for a failure-memory baseline.
- Research Intern at Multimedia Information Processing (MIP) Laboratory, Chungbuk National University, Korea (Mar. 2025 ~ Jun. 2026) / Advisor: Prof. Hyun Soo Kang / Conducted research on multimodal reinforcement learning for autonomous driving.
  - Designed and implemented RCS-AC, a CARLA driving actor-critic that gates RGB, LiDAR, route, and ego-state features by sensor reliability, risk, and Shapley contribution.

ACADEMIC ACTIVITIES & LEADERSHIP

- Researcher at Hello, World! Season 2 — World Models Paper Review, Pseudo Lab (가짜연구소), Korea (Oct. 2026 ~ Present) / Reviewing World Models research papers, participating in technical discussions, and contributing structured review documents to the project website.

- Advanced Division Leader at Applied Machine Learning Division, HyperCore AI & Data Analytics Club, Korea (Jan. 2026 ~ Sep. 2026) / Led machine learning study sessions and supported AI project development.

PUBLICATIONS & MANUSCRIPTS

Under Review

- Song, YongHwi* and Lee, Keon Myung
  "Planners Exploit Deterministic Error: Dithered Low-Bit World Models."
  Submitted to the Korean Artificial Intelligence Association (KAIA) Conference (poster), 2026.
  Extended version to be submitted to Robotics: Science and Systems (RSS) 2027.
- Song, YongHwi* and Lee, Keon Myung
  "Near-Miss Blindness: Latent World-Model Imagination Loses Resolution at the Outcome Boundary."
  Submitted to the Korean Artificial Intelligence Association (KAIA) Conference (poster), 2026.
  Extended version to be submitted to Transactions on Machine Learning Research (TMLR).
- Song, YongHwi* and Kang, Hyun Soo
  "Imagine Once, Act Fast: Lightweight Action Readout from a Frozen Driving World-Action Model."
  Submitted to the Korean Artificial Intelligence Association (KAIA) Conference (poster), 2026.

In Preparation

- Song, YongHwi*
  "K-PragShift: Diagnosing Korean Pragmatic Revisions in Tool-Using Language Models."
  To be submitted to the EACL 2027 Student Research Workshop.
- Song, YongHwi* and Lee, Keon Myung
  "Imagine Once, Act Fast: Lightweight Action Readout from a Frozen Driving World-Action Model."
  To be submitted to Transactions on Machine Learning Research (TMLR).

Conference Presentations

- Song, YongHwi* and Lee, Keon Myung
  "A World Model for LLM Agents: Comparing a Decision Model with General-Purpose LLMs."
  Oral presentation, 2026 Fall Conference of the Korean Institute of Intelligent Systems (KIIS).

  *First author.

PROJECTS

- Moonkeeper — CLPS Lunar Mission Explorer, 2026 NASA Space Apps Challenge Team Project, Korea / Building a React web app that visualizes CLPS lunar landing sites, Sun/Earth visibility windows, and mission timelines from NASA data. (Sep. 2026 ~ Present)
- Folding — Local-First Document Agent, Team Project, Korea / Developed a local-first document agent for retrieving, connecting, and safely editing heterogeneous documents. (Jul. 2026)
- Solar TutorBoard: AI-Powered Personalized Learning Platform, Upstage MixUp Agent Hackathon, Korea / Built AI-agent workflows and backend infrastructure for an educational AI platform. (May 2026)
- Vehicle and Pedestrian Trajectory Forecasting on Argoverse 2, Personal Project, Korea / Compared linear, LSTM, Transformer, and diffusion trajectory forecasters; PCA latent diffusion reached a minFDE of 0.86 at K=16. (Mar. 2026 ~ Jun. 2026)

COMPETITIONS & HACKATHONS

2026 NASA Space Apps Challenge — Seoul Local Event

- Accepted as a participant in the Seoul Local Event with Team Moonkeeper. (Nov. 2026, upcoming)

KRAFTON AI R&D Hackathon — Dream It Yourself, Video-Based Lightweight World-Action Model

- Participating individually in Track 3. Building a lightweight world-action model that infers hidden physics from pixels and plans pendulum swing-up in imagination. (Oct. 2026, in progress)

LG Aimers 9th Cohort — KBO Pitch Control Success Probability Prediction AI Hackathon

- Advanced to Phase 2 Online AI Hackathon (Top 20%). Worked on predicting pitch-control success probabilities from game context, player history, and tracking data. (Aug. – Sep. 2026)

2026 National Institute of Korean Language AI-Malpyung Competition — Argumentative Writing Scoring

- Ranked 23rd in the preliminary competition and advanced to the 50-team final arena. Developed an AI system for automated evaluation of Korean argumentative writing. (Jul. – Sep. 2026)

2026 CODEGATE AI Startup Hackathon — Finalist

- Selected as one of approximately 20 finalist teams for the offline hackathon and demo day. Participated in intensive development and validation of an AI-based startup MVP. (Jul. 2026)

Upstage × BDAI Agent Development Hackathon — Finalist

- Solar TutorBoard / Developed a Solar Pro3-based multi-agent platform for tutoring operations, including lesson reports, payment reminders, and schedule coordination. (2026)

AWARDS AND HONORS

- Academic Excellence Scholarship, Chungbuk National University, Korea (Mar. 2026, Mar. 2024, Sep. 2023)
- Undergraduate Research Assistant Scholarship, Chungbuk National University, Korea (Sep. 2025 ~ Mar. 2026)

ADDITIONAL PUBLICATIONS

1. Hyeban Namgung, Kuk Seorin, Park Geonwoo, Bae Seohyun, Song Yonghwi, Lee Hyesong, Han Jian, Hyojoong Kim, “A Comparative Study on the Prevalence of Forensic Flies Using Chicken Corpse”, Journal of Science Education for the Gifted, (2020)

SKILLS AND TECHNIQUES

- Programming:
  Python, C/C++, Java
- Machine Learning Frameworks:
  PyTorch, TensorFlow, scikit-learn, Stable-Baselines3
- World Models & Reinforcement Learning:
  World Models, World-Action Models, Model-Based Reinforcement Learning, Actor-Critic Methods
- LLMs & Agentic AI:
  Large/Small Language Models (LLMs/sLMs), Agentic AI, Tool-Using Language Models, Multi-Agent Systems
- Multimodal & Embodied AI:
  Multimodal Learning, Sensor Fusion, Computer Vision, Embodied AI, Physical AI
- Autonomous Driving & Simulation:
  CARLA, Argoverse 2, ManiSkill, Motion Forecasting, Autonomous Driving, Robotic Manipulation
- Development & AI Systems:
  Git, Linux, FastAPI, Supabase, Ollama, LM Studio, REST APIs